RMSD

The package `RMSD` contains a function named `RMSD`. It aims to provide a tool for multivariate outlier detection for elliptical data distribution based on the Modified Stahel-Donoho (MSD) estimators based on Béguin and Hulliger(2003).

The method requires a large number of projection as the data dimension increases, so please try the paralleled version, `RMSDp::RMSDp()` if the function `RMSD()` is terminated due to insufficient memory.

The function `RMSD::RMSD()` is faster than the function `RMSDP::RMSDp()` for lower dimensional datasets.

Arguments

The parameter 'nb' controls the number of basis for projection. Default setting is $\exp(2.1328 + 0.8023p)$ where p is the number of variables. Specifying a smaller number will reduce memory usage while it may degrade the performance of outlier detection.

The parameter `sd` is for a random seed. Plural execution without setting the random seed with a same dataset may yield different results, since the function `RMSD` uses random number.

The default threshold set to decide outlier is 99.9 percentile point of F-statistics. It can be changed using the parameter `pt`.

This function is an improved version of `msd` at https://github.com/kazwd2008/MSD by adding the last step to decide outliers by calculating the squared Mahalanobis distance of each observation and F-statistics to decide outliers. Its outlier detection performance is equivalent to that of the original function.

Value

- u : estimated robust mean vector
- V : estimated robust covariance matrix
- wt : final weights
- mah : squared Mahalanobis distnace of each observation
- FF : F test statistics
- cf : threshold to detect outliers (percentile point)
- ot : outlier flag (1:normal observation, 2:outlier)

Installation

from CRAN

The package RMSDP is on CRAN.

```
install.packages("RMSDps")
```

```
library(RMSDp)
```

from github

You can also install the package from GitHub with:

```
install.packages("devtools")
devtools::install_github("kazwd2008/RMSDp")
```

Example: Bushfire dataset [13 variables]

This is an example with the bushfire data set used by Campbell(1984) to locate bushfire scars.

The dataset contains 38 observations with 5 variables.

Usage

```
# install.packages("RMSD")
require(RMSD)

data(bushfire, package="robustbase")
o.msd <- RMSD(as.matrix(bushfire))

# scatterplot matrix
pairs(bushfire, pch=19, col=o.msd$ot)

# Q-Q plot
n <- nrow(bushfire)
d <- ncol(bushfire)
qqplot(qchisq(ppoints(n), df=d), o.msd$mah, pch=19, col=sort(o.msd$ot),
main = "Q-Q plot")
abline(0, 1, col = 'green')
```

Reference

Béguin, C. and B. Hulliger (2003) Robust Multivariate Outlier Detection and Imputation with Incomplete Survey Data, EUREDIT Deliverable D4/5.2.1/2 Part C.

Campbell, N. A.（1989）, Bushfire mapping using noaa avhrr data, Technical report, CSIRO